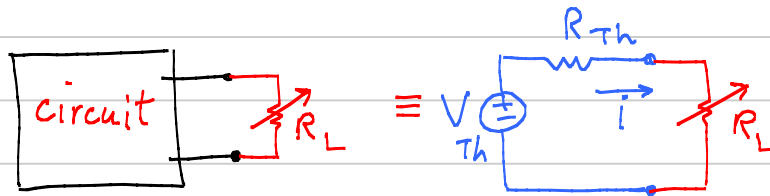




Maximum Power Transfer

To maximize the power transferred from a circuit to a load, we can replace the circuit by its Thevenin equivalent circuit and connect it to an adjustable load and find the load value for which the power is maximum.



$$P = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$

V_{Th} and R_{Th} are fixed. To find the value of R_L which maximizes the power, we need to differentiate P with respect to R_L and set it equal to zero.

$$\begin{aligned} \frac{dP}{dR_L} &= V_{Th}^2 \cdot \frac{d}{dR_L} \left[R_L (R_L + R_{Th})^{-2} \right] \\ &= V_{Th}^2 \left[\frac{1}{(R_L + R_{Th})^2} - \frac{2R_L}{(R_L + R_{Th})^3} \right] = 0 \\ \Rightarrow V_{Th}^2 \frac{R_L + R_{Th} - 2R_L}{(R_L + R_{Th})^3} &= 0 \end{aligned}$$

$$V_{Th} \neq 0 \Rightarrow R_{Th} - R_L = 0$$

$$\boxed{R_L = R_{Th}} \quad \text{for max power transfer}$$

under this condition ($R_L = R_{Th}$),

the maximum power transfer is:

$$P_{max} = R_L i^2 = R_L \left(\frac{V_{Th}}{2R_L} \right)^2$$

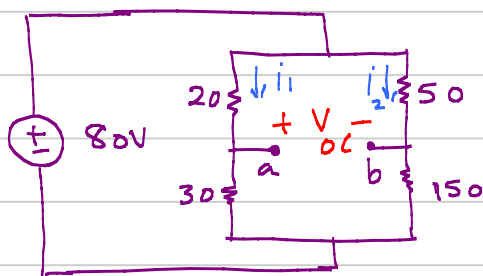
$$\boxed{P_{max} = \frac{V_{Th}^2}{4R_L}} \quad \text{maximum power}$$

Ex: Find R_L for maximum power transfer.
Find the maximum power.

First we need to find the

Thevenin equivalent circuit.

V_{Th} : $V_{Th} = V_{oc}$

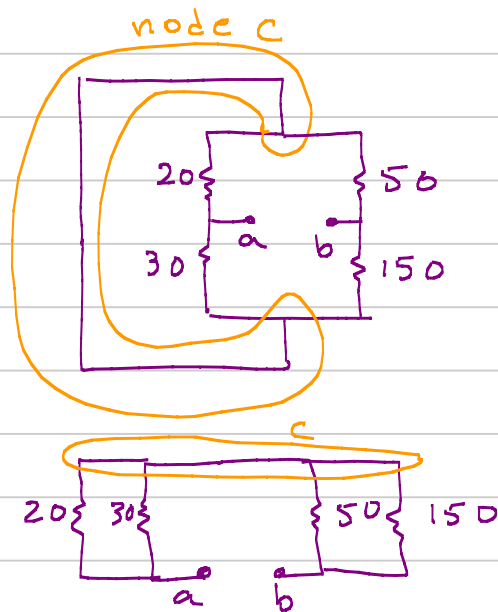


$$i_1 = \frac{80}{20+30} = 1.6 \text{ A}$$

$$i_2 = \frac{80}{200} = 0.4 \text{ A}$$

$$V_{oc} = -20i_1 + 50i_2 = -12 \text{ V}$$

R_{Th} : Turn off the source and find R_{eq} .



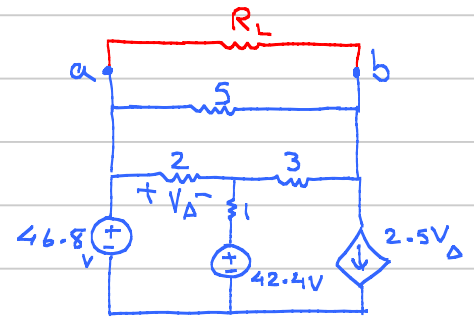
$$R_{ab} = (20 \parallel 30) + (50 \parallel 150) = \underline{49.5}$$

For max power transfer:

$$R_L = R_{Th} = \underline{49.5}$$

$$\begin{aligned} P_{max} &= \frac{V_{Th}^2}{4R_L} \\ &= \frac{(-12)^2}{4 \times 49.5} \\ &= \underline{0.73 \text{ W}} \end{aligned}$$

Ex: Find R_L for maximum power transfer. Find the maximum power.



$$\underline{V_{Th}}$$

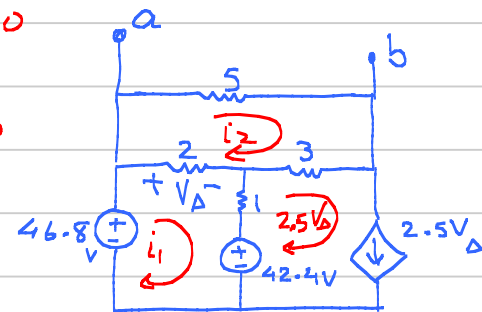
$$V_{Th} = V_{oc} = V_{ab}$$

$$\text{mesh 1 : } -46.8 + 2(i_1 - i_2) + 1[i_1 - 2.5(2)(i_1 - i_2)] + 42.4 = 0$$

$$\text{mesh 2 : } 5i_2 + 3[i_2 - 2.5(2)(i_1 - i_2)] + 2(i_2 - i_1) = 0$$

$$i_1 = 110, i_2 = 74.8$$

$$V_{Th} = V_{oc} = 5i_2 = \underline{374V}$$



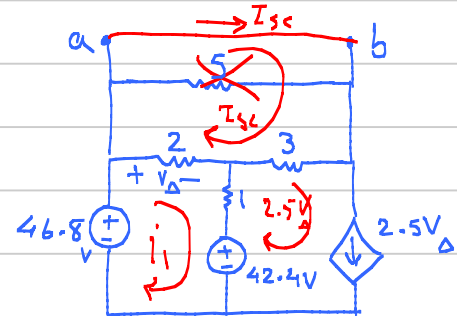
$$\underline{R_{Th}}:$$

$$3[i_{sc} - 2.5(2)(i_1 - i_{sc})] + 2(i_{sc} - i_1) = 0$$

$$-46.8 + 2(i_1 - i_{sc}) + 1(i_1 - 2.5(2)(i_1 - i_{sc})) + 42.4 = 0$$

$$i_{sc} = 6.8A$$

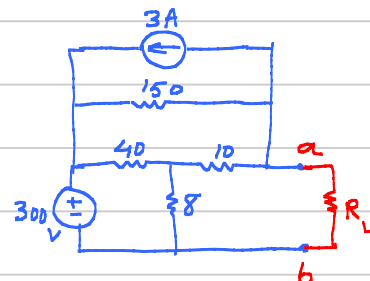
$$R_{Th} = \frac{V_{oc}}{i_{sc}} = \frac{374}{6.8} = \underline{55\Omega}$$



$$R_L = R_{Th} = \underline{55}$$

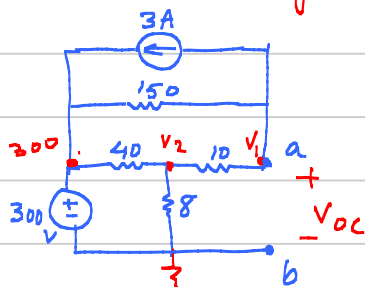
$$P_{max} = \frac{V_{Th}^2}{4R_L} = 635.8W$$

Ex: Find R_L for maximum power transfer. Find the maximum power.



V_{Th}

Nodal analysis



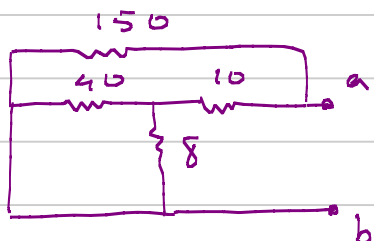
$$\frac{V_2 - 300}{40} + \frac{V_2}{8} + \frac{V_2 - V_1}{10} = 0$$

$$\frac{V_1 - V_2}{10} + \frac{V_1 - 300}{150} + 3 = 0$$

$$V_1 = 30V$$

$$V_{Th} = V_{OC} = V_1 = \boxed{30V}$$

R_{Th} :



$$R_{ab} = [(40 \parallel 8) + 10] \parallel 150$$

$$R_{Th} = R_{ab} = \boxed{15\Omega}$$

For max. power transfer $R_L = \underline{R_{Th} = 15\Omega}$

$$P_{max} = \frac{V_{Th}^2}{4R_L} = \underline{15W}$$

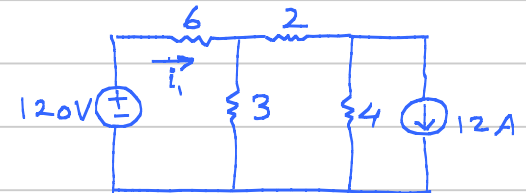
Principle of Superposition

A linear system obeys the principle of superposition, which states that whenever a system is excited by more than one independent sources, the total response is the sum of the individual responses.

Keep only one independent source at a time and turn off the remaining independent sources. Find the response to that source and repeat this for all independent sources. The total response is the sum of the responses.

A linear system is one whose output is linearly proportional to its input.

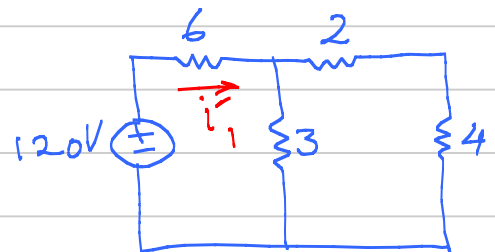
Ex. Find i_1 using superposition.



First, find the effect of 120V source on i_1 , by turning off all other sources.

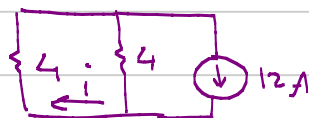
$$R_{eq} = 6 + [3 \parallel (2+4)] = 8 \Omega$$

$$i_1' = \frac{120}{8} = \underline{15A}$$



Then find the effect of 12A source on i_1 .

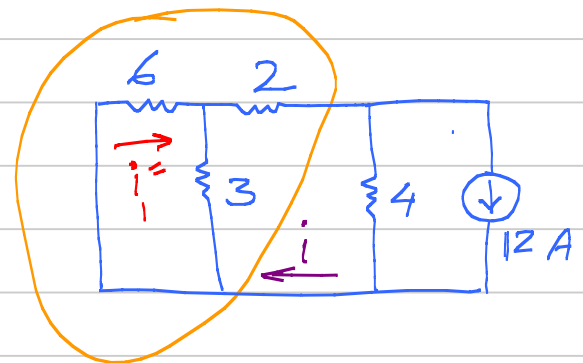
$$6 \parallel 3 + 2 = 4 \Omega$$



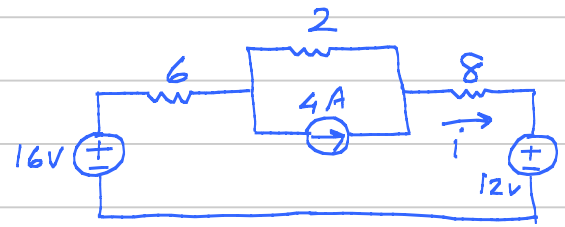
$$i = 12 \frac{4}{4+4} = 6A$$

$$i_1'' = i \frac{3}{3+6} = (6) \frac{3}{9} = \underline{2A}$$

$$i_1 = i_1' + i_1'' = 15 + 2 = \underline{17A}$$



Ex. Find i , using superposition.

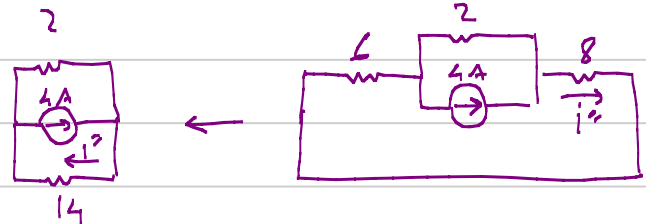


Consider the effect of independent sources one at a time.

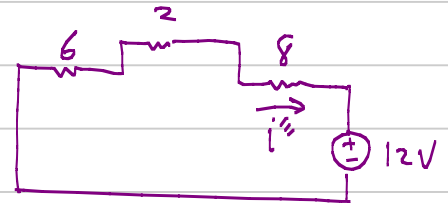
$$i_1 = \frac{16}{6+2+8} = 1A$$



$$i_2 = 4 \frac{2}{2+14} = 0.5A$$



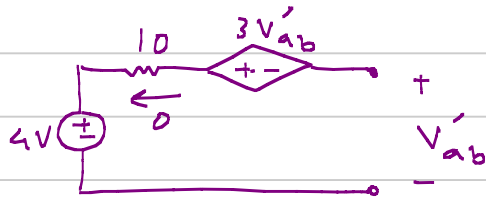
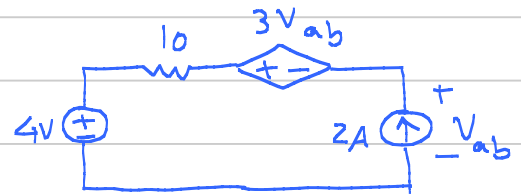
$$i_3 = -\frac{12}{6+2+8} = -.75$$



$$i = i_1 + i_2 + i_3 = 1 + .5 - .75$$

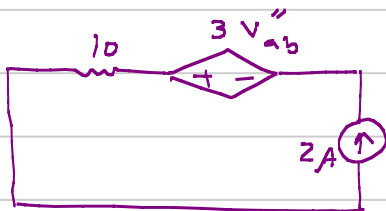
$$= \boxed{0.75} A$$

Ex. Find v_{ab} using superposition.



$$-V'_{ab} - 3V'_{ab} + 10(0) + 4 = 0$$

$$V'_{ab} = \underline{1V}$$



$$-V''_{ab} - 3V''_{ab} + 10(2) = 0$$

$$V''_{ab} = \underline{5V}$$

$$V_{ab} = V'_{ab} + V''_{ab} = \underline{6V}$$